
LECTURE NOTES

Lecture Notes

AI801 — Vidensrepræsentation

February 2026



Contents

Relations	3
Directed Graph	3
Reflexive, symmetrix, transitive properties	5
Functions	6
Forward image	6
Injective	6
Surjective	6
Bijection	7
Closure	7
The transitive closure of R	8
Composition	9

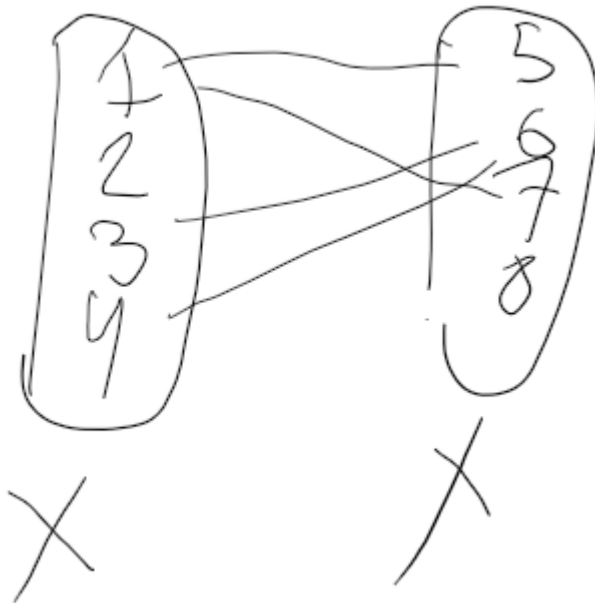
Relations

Definition

Given two sets X, Y a relation of type $X \times Y$ is just a set

$$R \subset X \times Y$$

if $(x, y) \in R$ we write xRy

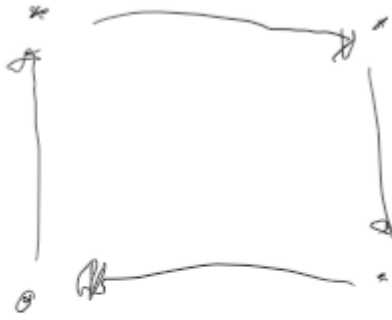


$$R = \{(1, 5), (2, 6), (3, 7), (4, 6)\}$$

Directed Graph

Graph

Def a directed graph (digraph) a pair $G = (V, E)$ Where E : Edges, V : Vertices We say E is a relation on V ($e_1 \dots e_n$) s.t. $e := (v_{i-1}, v_i)$

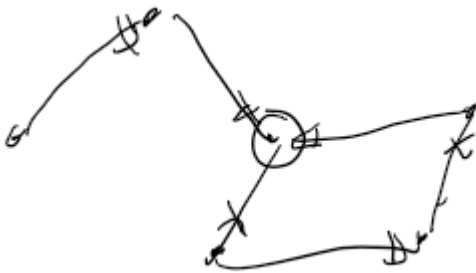


C_4



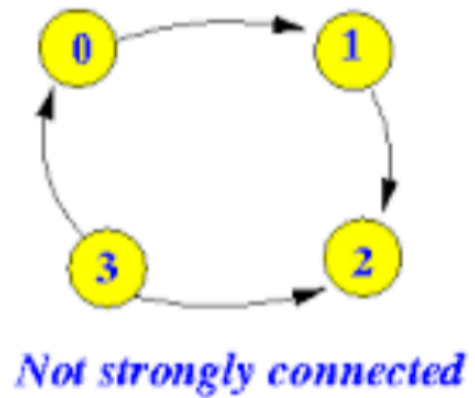
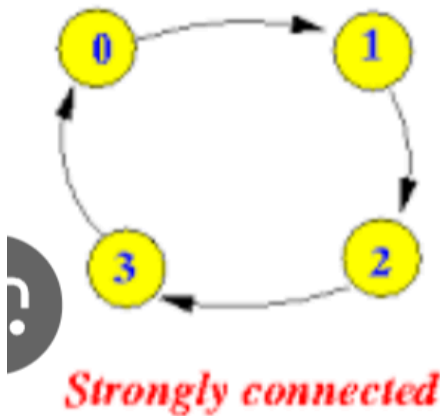
C_3

- If the edges is not distinsque way we call a walk



Strong connected diagraphs

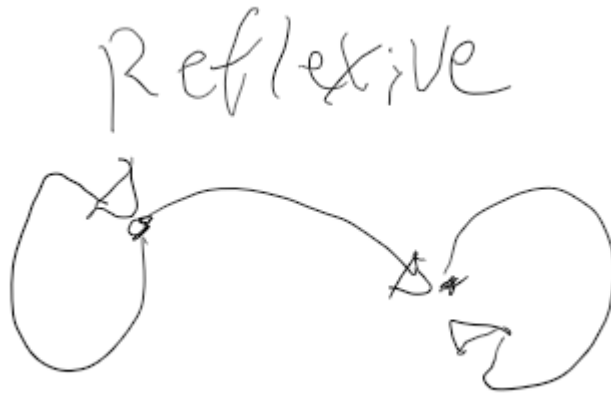
For all vertexes and edges if a vertexes path exist from v to u and from u to v . We called strongly connected



Reflexive, symmetric, transitive properties

- Reflexive iff for all $x \in X$ xRx

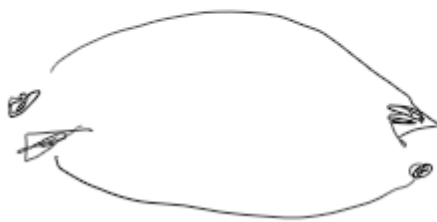
For reflexive $\leq \mathbb{N}$



- Symmetric iff for $x, y \in X$, $xRy \Rightarrow yRx$

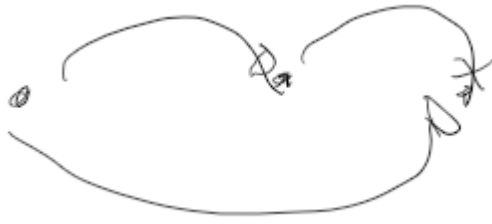
for symmetric $= \mathbb{Z} \mid x - y \mid = 1$

Symmetric:



- Transitive iff for $x, y, z \in X$ $xRy, yRz \Rightarrow xRz$

transitive $< \mathbb{N} \mid 1 < 2, 2 < 3, 1 < 3$



- If they all they have all equivalence relation

Functions

Functions

From x to y is a relation $f \subset X \times Y$

- $\forall x \in X, \exists y \in Y$ s.t. xfy
- $\forall x \in X$ the forward image $f(x)$ has a single element.

Forward image

$$R \subseteq X \times Y, P \subseteq X \text{ of } P :$$

Is the set $Q = \{y \in Y \mid \exists x \in P\}$

- The forward image is simply a subset of $R = P$ and then follows the relations from P to Y .

$$P_o = \{10, 20\}$$

$$Q_o = R(p_o) = \{5, 6\}$$

$$p_1 = \{30, 40\}$$

$$Q_1 R(p_1) = \{7\}$$

Injective

Definition

- A function iff. $\forall x_1, x_2 \in X \ f(x_1) = f(x_2) \Rightarrow x_1 = x_2$

A function f is said to be one- to- one, or a injection, if and only if $f(a) = f(b) \Rightarrow a = b$ in the domain. Which means for a function value have the same out given two inputs value, the input values has to be the same value.

- $f(x) = x^2$ is not injective, since we can find two inputs in the domain $\forall x \in X, \mathbb{R} \mid f(x) = f(-x)$ since $x^2 = (-x)^2$

Surjective

Definition

$$\forall x \in X, \exists y \in Y, f(x) = y$$

$f : \mathbb{Z} \rightarrow \mathbb{Z}x \Rightarrow x^2$ Forevery element $b \in B$ in the codomain, has at least one $a \in A$ in the domain that maps $f(a) = b$

$$\forall b \in B, \exists a \in A \mid f(a) = b$$

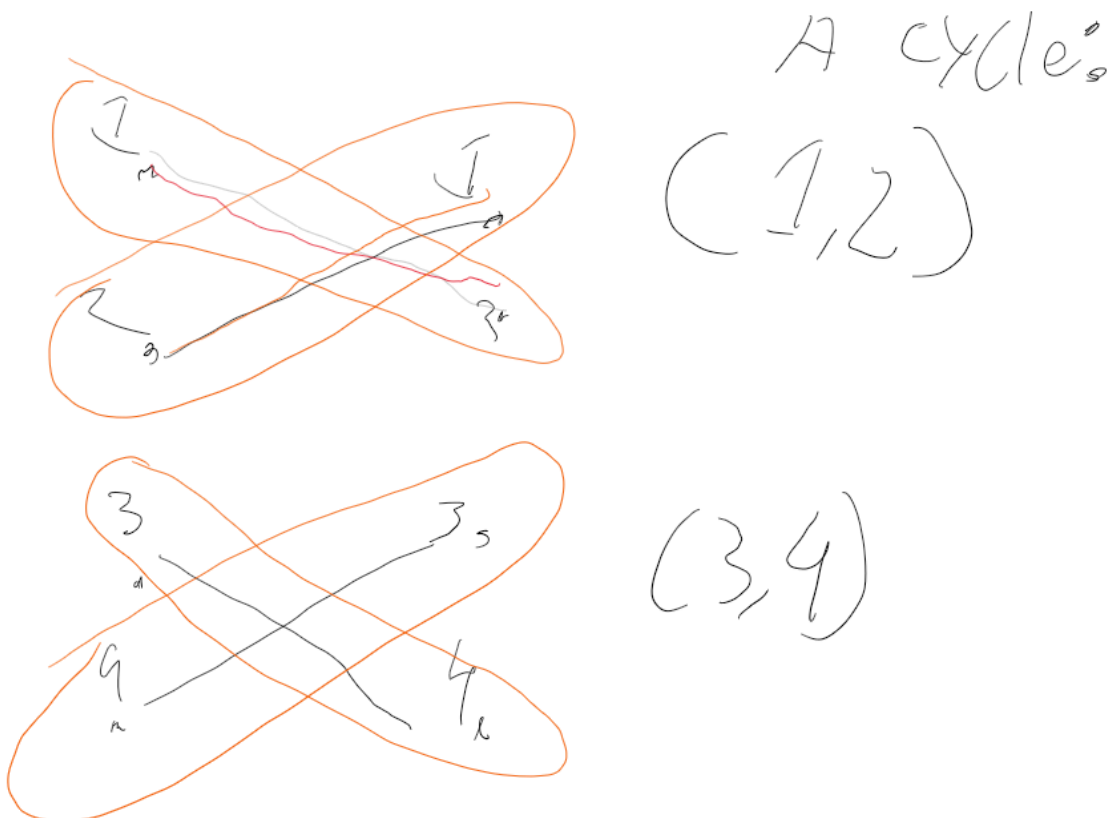
Bijection

Definition

A bijection $f : X \rightarrow X$ with X fininte is called a permutation. The function f is one-to-one correspondence, or a bijection, if it its both injective and surjective

e.g. $f(x) = 2x + 1$ is bijective

- A cycle is a a function that maps elements in a set in a circular, ordered fashion, where each elemetns maps tp the next and the last maps back to the first



This cycle contains (1,2)

Closure

Definition

If R is a relation on a set A , then the closure of R with respect to P , if it exists, is the relation S on A with property P that contains R and is a subset of every subset of $A \times A$ containing R with property P

s.t. "the closure is the smallest relation that contains R and satisfies property P ."

$$R \subset X \times X$$

The transitive closure of R

The smallest relation is of R with is transitive. Extend: consider relation which include the relation. Let R be a relation on a set A . The connectivity relation R^* consists of the pair (a,b) such that there is a path of length at least one from a to b in R .

One can say that finding the transitive closure of a relation is equivalent to determining which pairs of vertices in the associated graph are connected by a path.



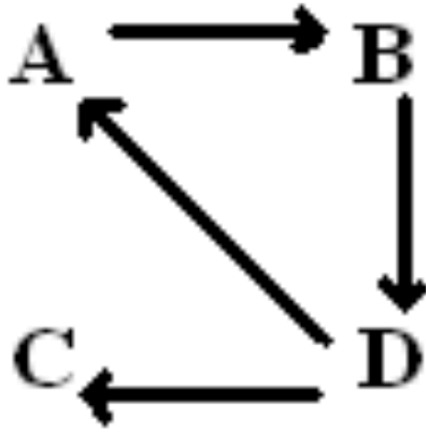
Connectivity relation R^*

Let R be a relation on a set A . The connectivity Relation R^* consists of the pairs (a, b) such that there is a path of length at least one from a to b in R . Since R^* consists of the pairs (a,b) such that there is a path of length n from a to b , it follows that R^* is the union of all the sets R^n :

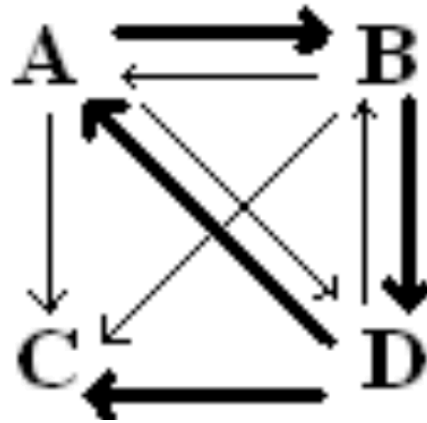
$$R^* = \bigcup_{n=1}^{\infty} R^n$$

- Rules of thumbs R^* just means if a path exist such a path is from a to b . R^1 they are directly connected aka. a path of length 1, R^2 is a path of length of 2. Remember this is for directed connections since pairs are ordered structures.

Original



Transitive Closure



e.g.: with the relation R

$$R = \{(x, y), (y, z)\}$$

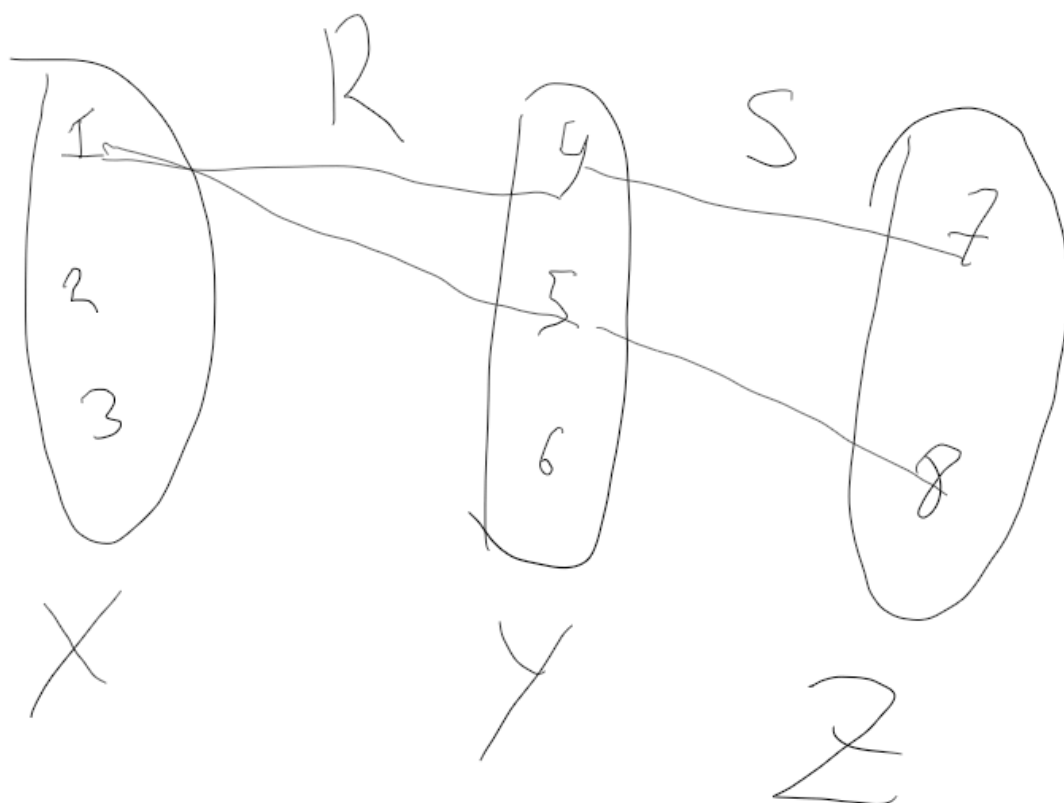
The positive transitive closure, is the closure that satisfies the property p.

$$R^+ = \{(x, y), (y, z), (x, z)\}$$

Composition

Def $R \subseteq X \times Y, s \subseteq Y \times Z$ The composition relation: $S \circ R$ is the set $S \circ R = \{(x, y) \in X \times Z \mid \exists y \in Y, xRy \wedge ySz\}$

e.g.



$$R = \{(1, 4), (1, 5)\} = xRy$$

$$S = \{(4, 7), (5, 8)\} = ySz$$

$$s \circ r = \{(1, 7), (1, 8)\}$$

the composition $R \subseteq XxY$

$$R \circ R = R^2$$

$$\underbrace{R \circ R \circ \dots \circ R}_{n \text{ times}} = R^n$$